Regression Diagnostic

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#Regression Diagnostic Paper

Introduction To complete this assignment properly I examined the difficult but very cool model I created to predict current vote share. The independent variables were media mentions, campaign spending, and previous vote share.The goal I was trying to achieve is to see which factors seem to influence current electoral performance. I discovered that the model used OLS regression I felt it was important to see if the model violated any assumptions. My research led me to understand that it is very important to see if the residuals appeared random and whether the residuals appeared random. Also, I noted that it is important to test if the variance of the errors remained consistant across the model. My research also led me to discover that the best way to do this was to run a heteroskedasticity diagnostic.A heteroskedasticity diagnostic tests whether the errors in a regression model remain consistent across the data. In regression analysis, the model assumes that the errors have roughly the same amount of variation throughout the dataset. If the variation changes too much, the standard errors and significance tests can become unreliable. This was the best diagnostic choice because the project uses OLS regression, and constant error variance is one of the main assumptions of that type of model. Testing for heteroskedasticity helps determine whether the regression results can be interpreted with confidence. Yay me for figuring this out! Library

(lmtest) library(sandwich)

Loading the libraries I need and the data

```
model1 <- lm(current_vote ~ media_mentions + spending + previous_vote, data = x)
```

```
summary(model1)
```

Regression Results The regression results showed that previous vote share predicted current vote share better than any other variable in the model. Candidates or parties that performed well in earlier elections usually performed well again in the current election. Previous vote share showed the strongest relationship with election results and remained statistically significant throughout the analysis. This is my take, and I hope I interpreted this correctly. Based on the research I did in how to view these studies I think I am correct: Media mentions and campaign spending both showed small positive relationships with current vote share, but neither one showed a strong enough effect to confidently say they influenced election results on their own. Once I included previous vote share in the model, earlier electoral success explained most of the results. The model explained about 60 percent of the election results. That is a pretty strong result because many different things can affect how people vote.

Diagnostic Test

r

plot(model1, which = 1)

You can look at the plot I included in my email to see this visually. Most of the points stayed fairly close to zero and did not show any major extreme pattern, which suggests the model fits the data reasonably well. The red line was not perfectly straight though. It dipped slightly and then went back up, which based on what I looked up could mean the relationships in the model are not perfectly linear. I also noticed that the residuals spread out a little more at higher fitted values. Based on what I looked up, this may show mild heteroskedasticity, which means the amount of error changes across the data. The pattern did not look severe, but it still gave me a reason to run a heteroskedasticity test and use robust standard errors.

R

bptest(model1) coeftest(model1, vcov = vcovHC(model1, type = “HC1”))

The diagnostic results did not look serious enough to make the model unusable. Even after checking for possible heteroskedasticity, the overall results stayed mostly the same. Previous vote share still predicted current vote share better than the other variables, while media mentions and spending still did not show strong statistical significance.I also noticed a few points that sat farther away from the main group of residuals. Based on what I looked up, these might represent unusual elections or candidates that differed from most of the sample. Even so, they did not seem extreme enough to seriously affect the overall model.

Conclusion

The diagnostic tests suggest the regression model works fairly well. I found some signs of mild nonlinearity and possible heteroskedasticity, but based on what I looked up, neither issue seemed serious enough to ruin the model. Using robust standard errors also seemed to help account for some of those possible problems and made the results feel more reliable. The biggest finding from the model was that previous electoral performance strongly predicted current electoral performance. Media mentions and campaign spending may still matter in real elections, but this model did not show much statistical evidence that either one independently affected vote share once previous electoral success was included.